

**In one line:** ARC = the compiler inserts retain/release; an object is freed *deterministically* the instant its **strong** count hits 0 (deinit runs). No tracing GC ⇒ **no cycle detection**: two objects that strongly own each other leak forever, silently. You break cycles by making the back-edge weak or unowned.

## How it works

- ARC manages **class instances** (+ actors, closure contexts). Structs/enums are copied, not counted — but copying a struct that holds a class ref retains that object.
- strong** (default) — owns: +1. **weak** — non-owning, must be var + Optional; the runtime (side table) sets it to nil when the object dies. **unowned** — non-owning, not auto-nilled; access after death = **crash** (unowned(unsafe) = dangling pointer, UB).
- Ownership ≠ nil-ability**. Two independent dials: ? answers “can it be nil?”; strong/weak/unowned answers “does it keep the object alive?”. var x: Foo? retains exactly as hard as var x: Foo.
- weak vs unowned**: can the referent die while I still hold the ref? *Maybe* → weak (delegates, async callbacks — the safe default). *Never* — it outlives me → unowned (child that cannot exist without its parent: in CreditCard, unowned let customer: Customer).
- Cycle shapes**: (1) **delegate** — make the protocol: AnyObject and the property weak var delegate; (2) **closure** stored on self that captures self (completion handlers, lazy var, Combine sink); (3) **parent/child back-refs** — parent owns children strongly, child.parent, list.prev are weak; (4) **Timer** — the run loop retains it: also invalidate().
- Capture list [x]** = a **by-value snapshot taken when the closure is created**. Without a list the closure captures the *variable* (sees later writes). For a **reference type** the snapshot copies the *pointer*: same object, so later mutations of the object are visible. [weak self] is the same snapshot, just non-owning. Mix: [weak self, count].
- Only **escaping** closures stored (directly or transitively) on self need [weak self]; map/forEach are non-escaping.

## Example — break the cycle, prove it

```
final class VM {
    var onChange: (() -> Void)? // VM owns the closure
    func bind() {
        onChange = { [weak self] in // closure does NOT own VM
            guard let self else { return } // unwrap once
            self.reload() }
    }
    func reload() {}
}

func test_VM_deallocates() {
    var sut: VM? = VM(); sut?.bind()
    weak var weakSut = sut // observer, not an owner
    sut = nil // drop the last strong ref
    XCTAssertNil(weakSut, "VM leaked - retain cycle?")
}
```

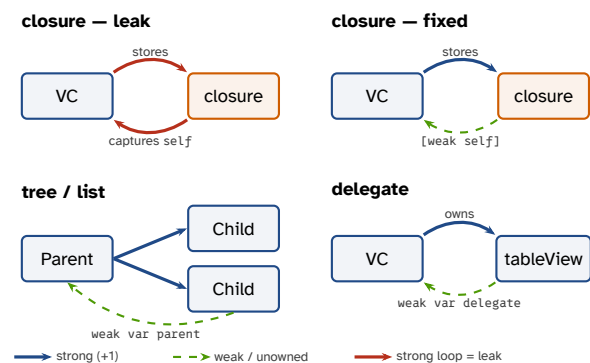
## Memory tools

- Xcode Memory Graph Debugger** (debug-bar button): live objects + *who holds whom*; purple ! = leak/cycle. Turn on *Malloc Stack Logging* to see where it was allocated.
- Instruments — Leaks**: finds *unreachable* blocks (true leaks, cycles).
- Instruments — Allocations**: heap growth over time; *Mark Generation* exposes *abandoned* memory (reachable but never freed, e.g. an unbounded cache).
- Zombies**: use-after-free / EXC\_BAD\_ACCESS.

## The two dials

|         | Foo (non-optional)                 | Foo? (optional)                                  |                        |
|---------|------------------------------------|--------------------------------------------------|------------------------|
| strong  | owns (+1)<br>never nil             | still owns (+1)<br>may be nil — <b>not weak!</b> |                        |
| weak    | compile error<br>(weak must be ?)  | non-owning<br><b>auto-nil</b> on dealloc         | ? = can be nil         |
| unowned | non-owning<br><b>crash</b> if dead | non-owning (Swift 5+)<br>not auto-nilled         | keyword = keeps alive? |

## Cycles and their fix



## Traps

- ! “Optional means non-owning” — no: var parent: Node? is *strong*; write weak.
- ! [x] is **by value**, not by reference: snapshot at creation. Reference type → the *pointer* is copied.
- ! Proving no leak: weak ref → set strong to nil → XCTAssertNil. A deinit print is only the manual version.
- ! “No leaks” but memory grows → abandoned memory → Allocations.
- ! [unowned self] in a network callback that outlives the VC = crash.
- ! Timer + [weak self]: the run loop still holds the timer — invalidate().

## self in closures — what each Swift allows

| Version | Feature                                            | Code Example                |
|---------|----------------------------------------------------|-----------------------------|
| 5.0     | unowned references may be Optional                 | unowned var x: T?           |
| 5.3     | SE-0269: implicit self after [self] or in a struct | { [self] in reload() }      |
| 5.7     | SE-0345: if let shorthand                          | guard let self else { ... } |
| 5.8     | SE-0365: implicit self once unwrapped              | [weak self] ... reload()    |

## Remember

“?” asks *may it be nil*; the keyword asks *do I keep it alive*. Arrows point *down* the tree strong, *back up* weak. Tools: Graph, Leaks, Allocations — “GLA”.